

RenSeg: Leveraging Unsupervised Segmentation using Localization and Contour-Guided Quickshift for Renal Calculi and Carcinoma Segmentation and Classification

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Abstract—Chronic kidney diseases present a major global health challenge, often leading to renal failure due to delayed diagnosis and the shortage of nephrologists, which limits timely treatment. Therefore, the development of AI-driven systems for early detection is essential. However, most existing automated renal disease detection methods rely on supervised learning frameworks that require large amounts of clinically annotated data, which are often scarce and difficult to obtain. To address these challenges, this study proposed kidney localization and an unsupervised contour-guided quickshift-based automated segmentation method namely RenSeg on a dataset of 8,737 axial and coronal CT-scan images. RenSeg focuses on two critical diseases: Renal Calculi, which are highly prevalent and associated with recurrent pain and long-term kidney dysfunction, and Renal Carcinoma, a life-threatening tumor if not detected early. The analysis shows that unsupervised segmented images outperform manually annotated datasets. RenSeg achieved Dice scores of 0.9458 for calculi and 0.9309 for carcinoma, a precision of 0.95 and a recall of 0.94. MobileNetV2 reached the highest classification accuracy of 0.98 on RenSeg compared to 0.92 on manual annotations. The proposed unsupervised RenSeg approach outperformed the manually annotated approach across all three models in terms of accuracy, F1-score, and ROC-AUC, demonstrating superior generalization. This research demonstrates that optimizing the region of interest (ROI) enhances predictive accuracy while reducing annotation effort, making RenSeg a scalable solution for timely renal disease detection.

Index Terms—Calculi, Carcinoma, Classification, IoU, Renal, Segmentation.

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I. INTRODUCTION

Chronic kidney disease (CKD) is a growing global public health issue, currently affecting over 10% of the global population, and is projected to become the fifth leading cause of death worldwide by 2040 [1]. Kidney diseases commonly include renal calculi (stones) and renal carcinoma (tumors). Renal calculi disease affects the human body's fluid balance, blood purification, and blood pressure. In fact, 70% of the kidney disease cases are associated with diabetes and hypertension [2]. Although various imaging modalities, such as CT scans, X-rays and MRI are used to diagnose renal conditions, CT scans, which utilize X-ray beams to generate cross-sectional views, remain the most effective for providing detailed 3D anatomical information. However, in South Asia, diagnosis is limited by a severe shortage of specialists, with only one nephrologist per million people compared to 25.3 per million in Europe [3].

Despite progress in medical imaging, most automated renal disease detection methods depend on supervised learning frameworks that require extensive clinically annotated data [4]. This reliance on nephrologist labeling and inclusion of irrelevant regions limits both scalability and accuracy. To overcome this issue, this study proposes an unsupervised segmentation-based classification method using CT-scan images, as illustrated in the graphical abstract (Fig. 1). It combines kidney localization and unsupervised segmentation to isolate the region of interest (ROI), focusing exclusively on kidney structures. This approach minimizes annotation and computational requirements while enabling accurate classification of renal calculi and carcinoma with lightweight deep learning models. It is suitable for deployment in resource-limited clinical settings without compromising diagnostic accuracy. The major contributions of this research are as follows:

- We propose a kidney localization approach on CT-scan images that limits analysis to the region of interest (ROI), demonstrating that excluding non-relevant regions improves classification performance.
- We explore the unique relationship between Aorta vessel characteristics in CT-scan images of calculi-infected, carcinoma-infected, and normal kidneys.

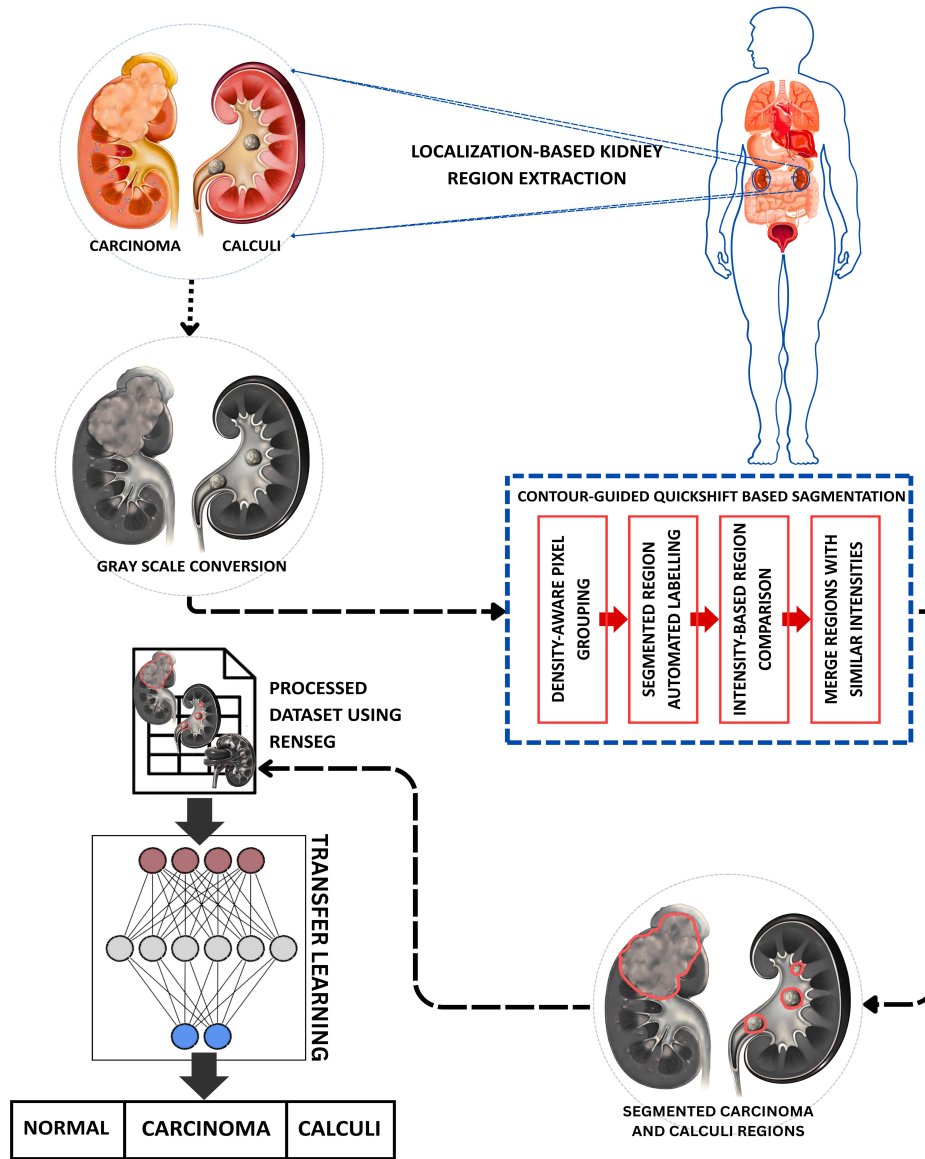


Fig. 1: Schematic overview of the proposed RenSeg system. The system begins with CT-scan images, undergoes preprocessing, followed by region of interest (ROI) extraction to isolate kidney structures. These processed images are automatically labeled using the C-G Quickshift method and then used to train a transfer learning-based network for classification. The output is the identification of kidney anomalies, distinguishing among normal kidneys, renal calculi, and renal carcinomas.

- We introduce a modified unsupervised segmentation technique, termed Contour-Guided Quickshift, which mitigates over-clustering and enhances boundary precision.
- We develop a manually annotated CT dataset of renal calculi and carcinoma under the direct supervision of a certified nephrologist of a renowned medical institution, following all necessary ethical permissions, providing a valuable benchmark for future studies.

The paper is organized as follows: Section II covers the preliminary study, Section III reviews related literature, Section IV describes the methodology, Section V presents results analysis, Section VI discusses the findings, Section VII concludes the paper with a discussion of limitations and future work, and Section VIII provides data and code availability.

II. PRELIMINARY

This section provides a concise overview of the normal kidney as well as pathological conditions such as renal calculi and carcinoma, covering their classification, severity, and diagnostic complexities.

A. Normal Kidney

The kidney is a key organ in the urinary system, essential for maintaining the body's internal balance. It filters blood, removes metabolic waste, and regulates fluid and electrolyte levels [5]. Each kidney contains roughly one million nephrons, with the renal cortex and medulla coordinating blood filtration and urine production, while the renal pelvis directs urine to the bladder (Fig. 2(a)). It also controls blood pressure

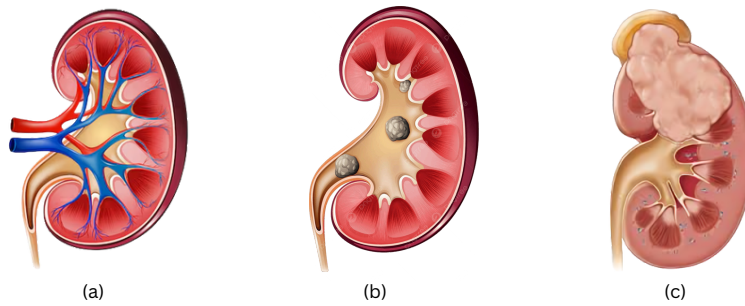


Fig. 2: Comparative anatomical illustration of the kidney demonstrating three distinct conditions: (a) normal, (b) stones, (c) tumor.

through the renin, angiotensin, and aldosterone system, stimulates red blood cell production by releasing erythropoietin, and maintains acid-base balance through specialized tubular mechanisms [6], [7]. Disruption of this normal physiology, for example by nephrolithiasis or carcinoma, can significantly impair kidney function and lead to systemic effects [5].

B. Renal Calculi

Renal calculi have been a very common urinary tract problem for such a long time (Fig. 2(b)). Preventing recurrence of the disease is a significant concern for public health since it is closely associated with an increased susceptibility to long-term kidney dysfunction, kidney malfunction, elevated blood pressure, cardiovascular disorders, and metabolic disorders such as diabetes [8]. Kidney dysfunction refers to impaired renal function, in which the kidneys continue to function but at a reduced efficiency. In contrast, kidney malfunction implies the failure of the organ to function properly, often indicating a sudden or complete breakdown of its normal physiological role. Moreover, renal calculi exhibit variation in morphology, chemical composition, and size, primarily due to abnormalities in urinary solute concentrations. Clinically, they are classified into five types based on mineral content: calcium-based, struvite (ammonium magnesium phosphate), uric acid, cystine, and drug-induced calculi [9].

C. Renal Carcinoma

Renal carcinoma is an aberrant proliferation of cells within the kidney, which can potentially be life-threatening (Fig. 2(c)) [10]. Abnormal DNA mutations in renal cells are the primary cause of carcinoma development, leading to uncontrolled cell proliferation and potential metastasis to other organs [11]. Moreover, it can develop within the kidney or spread to organs such as the lungs. Annually, renal carcinoma affects more than 400,000 individuals worldwide, representing a significant health risk [10]. Renal carcinoma includes various malignant and benign growths, originating from different nephron areas and influenced by genetic and clinical factors. Benign forms generally grow slowly, rarely metastasize, and have a low risk of recurrence. However, some benign forms can become malignant, which emphasizes the importance of timely treatment [12].

III. LITERATURE REVIEW

Vedaldi and Soatto [13] proposed the Quick Shift algorithm, a non-iterative and density-based clustering method designed for mode seeking in high-dimensional data. Unlike mean shift, which iteratively shifts each data point toward the nearest high-density region, or medoid shift, which updates cluster centers to the most centrally located data point within a neighborhood, Quick Shift algorithm connects each sample directly to its nearest neighbor with higher estimated density and introduces a tunable threshold to balance under and over fragmentation, making it both simple and effective for various image segmentation tasks. In our work, we introduce a modified Quick Shift tailored for renal CT segmentation to improve accuracy and reduce fragmentation.

Islam, M.N. et al. [14] worked on six models from Vision Transformers and deep learning models. They found the Swin transformer is not only more accurate with 0.993 but also trains more quickly than others. The VGG16 model is better than ResNet50 and Inceptionv3 by exhibiting the necessary anatomical defects in detail, according to the study's attempt to uncover the black box of these models. This indicates the importance of model selection, while our work demonstrates that unsupervised segmentation can further enhance classification performance.

Moreover, H. Cui et al. [15] proposed a hybrid U-shaped model for kidney segmentation in polycystic kidney disease. The approach integrates wavelet pooling, cascaded residuals, and efficient multi-head self-attention to enhance feature extraction and segmentation accuracy. Tested on MRI images, it achieved a Dice score of 91.5%, closely matching manual results, significantly reducing processing time, and enabling precise volume measurement. Yet, the method is modality-specific (MRI), whereas our study focuses on CT scans.

De Boisredon d'Assier et al. [16] proposed MoDATTS, a two-stage method for tumor segmentation across image modalities. The first stage uses a CycleGAN model to translate images from one modality to another while preserving tumor structures. In the second stage, a segmentation model is trained using synthetic and real unlabeled images with semi-supervised learning. Iterative self-training generates pseudo-labels on unlabeled data, improving segmentation accuracy without extensive annotations. However, this approach still relies on pseudo-labeling, which can limit scalability in fully unsupervised settings like ours.

Similarly, Wu et al. [18] proposed a dense object detection

TABLE I: Comparison of prior ML/DL methods and RenSeg for renal disease diagnosis across imaging modalities, summarizing objectives, models, accuracy, strengths, and limitations.

Reference	Class	Objective	Model	Acc.	Strengths	Limitations
H. Cui et al. [15]	Kidney	Supervised Segmentation	Hybrid U-Net	0.915	Accurate MRI segmentation; improved feature extraction.	MRI only; lacks classification; not unsupervised.
De Boisredon d'Assier et al. [16]	Tumor	Semi-supervised segmentation	MoDATTS	0.92	Cross-modality tumor segmentation; semi-supervised learning.	Needs pseudo-labels; not applied to kidney CT.
Bayari et al. [17]	Kidney Stone	Secure segmentation in IoMT	Watermarking + CNN	0.96	Ensures IoMT data security; good segmentation accuracy.	High computation; kidney stone only; lacks classification.
Islam, M.N. [14]	Cyst, Stone & Tumor	Supervised segmented classification	Swin Transformer	0.99	High accuracy; detailed feature extraction.	Requires extensive annotation; high computational cost; not unsupervised.
RenSeg (Proposed)	Calculi & Carcinoma	Unsupervised segmented classification	RenSeg+ResNet50	0.95	Fully unsupervised segmentation and classification; reduces annotation time; scalable to CT dataset.	Limited to renal calculi and carcinoma.
			RenSeg+VGG19	0.97		
			RenSeg+MobileNetV2	0.98		

framework named D-YOLOv8, built upon the YOLOv8 architecture, to address the challenges of detecting small, densely packed targets from UAV imagery. From their findings, D-YOLOv8 integrates Dense Feature Pyramid Networks (D-FPN) and a Dense Attention Layer (DAL) to enhance feature extraction and focus on relevant dense regions. The model achieved a high average precision (AP) of 98.37% with only 13.2 million parameters, outperforming conventional YOLO models in both detection accuracy and computational efficiency. This highlights the potential of YOLOv8 for precise kidney localization in our RenSeg framework.

Elmessery et al. [19] presented a SegFormer-based semantic segmentation framework for microbial alterations in strawberry diseases. The study evaluated three Mix Transformer encoders (MiT-B0, MiT-B3, and MiT-B5), demonstrating that transformer-based architectures can achieve superior segmentation accuracy and efficiency compared to conventional convolutional approaches. Using the MiT-B0 encoder as the proposed configuration, their framework achieved a mean IoU of 94.37% and a mean pixel accuracy of 96.82%, underscoring the effectiveness of transformer-based models for precise and efficient segmentation. This shows the potential of transformer-based architectures, although our approach focuses on lightweight CNNs combined with unsupervised segmentation.

Eliwa et al. [20] proposed a secure and transparent lung and colon cancer classification framework by integrating blockchain with Microsoft Azure cloud services. Using a decision tree classifier as the proposed model, their approach achieved an accuracy of 98.3%. Their study emphasized the importance of combining accurate cancer classification with blockchain-based data management to ensure transparency, integrity, and security in medical diagnostics. While their focus is on secure classification, our study emphasizes unsupervised segmentation accuracy and efficiency.

Furthermore, Mostafa et al. [21] proposed a feature reduction framework for hepatocellular carcinoma prediction using multiple machine learning algorithms. Their approach employed feature selection and dimensionality reduction techniques, with the random forest classifier achieving an accuracy of 97.6% on the clinical dataset. Unlike their work with structured clinical data, our study addresses CT-scan based unsupervised segmentation and classification of renal calculi

and carcinoma.

Table I has been compiled to evaluate the effectiveness of machine learning (ML) and deep learning (DL) approaches in renal disease detection, including the proposed RenSeg model.

IV. METHODOLOGY

This research processes CT-scan images using two approaches for classification. In the first, manual annotation under a nephrologist's supervision produces a fully annotated dataset. In the second step, the same raw images undergo localization, followed by the RenSeg method, which incorporates the contour-guided Quickshift segmentation algorithm as a key component, both summarized in Algorithm 1 and Algorithm 2. In both approaches, images are cropped, resized, and split into training, validation, and testing sets, with data augmentation applied to training sets to address class imbalance. Finally, transfer learning models such as ResNet50, VGG19, and MobileNetV2 are used to evaluate the effectiveness of both approaches. This comparative study aims to determine whether the RenSeg-based approach can match or surpass conventional manual annotation in downstream classification tasks.

A. Data Description

The CT kidney dataset - Normal, Tumor, Cyst, and Stone by Nazmul Islam [14] is an important resource for kidney health research. It comprises Digital Imaging and Communications in Medicine (DICOM) images, which are the standard format for storing, transmitting, and visualizing medical imaging data, illustrating normal kidney anatomy as well as various pathological conditions, including carcinoma, cysts, and calculi. This diversity supports a broad range of machine learning applications such as classification, segmentation, and feature extraction for computer-aided diagnosis. In this study, only the Normal, Calculi, and Carcinoma classes were utilized, comprising a total of 8,737 CT-scan images, as shown in Table II. The research specifically concentrated on renal calculi and carcinoma, which are among the most prevalent and clinically significant renal diseases. By narrowing the scope to these two conditions, this work aims to develop an unsupervised approach capable of improving classification and diagnostic support for the most common kidney pathologies encountered in clinical practice.

TABLE II: Dataset Summary

Class	Number of Images
Normal	5077
Calculi	1377
Carcinoma	2283
Total	8737

TABLE III: Dataset composition after annotation

Class	Original	Annotated
Renal Calculi	1,377	Yes
Renal Carcinoma	2,283	Yes
Normal	5,077	No
Total	8,737	3,660

B. Manual Annotation

As a comparative study, this research investigates the effectiveness of supervised segmented CT-scan images in lightweight classification models. Using the publicly available CT kidney dataset by Nazmul Islam [14], a total of 1,377 CT-scan images depicting renal calculi and 2,283 CT-scan images depicting renal carcinoma were manually annotated to highlight the regions of interest, as shown in Fig. 3. These annotations were performed under the direct supervision of a certified nephrologist (kidney specialist) from a renowned medical institution, following all necessary ethical permissions. This annotation process resulted in the creation of a new dataset consisting of 3,660 supervised segmented images. In contrast, 5,077 CT-scan images labeled as Normal were retained from the original dataset without further annotation, as they did not exhibit any abnormalities requiring segmentation. The detailed dataset composition is summarized in Table III. This combined dataset, which includes annotated renal calculi and carcinoma images as well as non-annotated normal kidney images, was utilized to evaluate classification performance under different settings.

C. Proposed Unsupervised RenSeg Method

This section outlines a novel unsupervised RenSeg approach comprising three components: kidney localization for region extraction, contour-guided Quickshift segmentation for precise boundary detection, and data resizing for uniform model performance. The overall method is illustrated in the top-level overview (Fig. 4), while the detailed RenSeg model architecture is shown in Fig. 5. Together, these components form a novel and effective model for unsupervised medical image segmentation, and each phase is presented in Fig. 6.

1) Kidney Localization: RenSeg employed YOLOv8-m for object localization on CT-scan images. A 20% subset of 1,748 images from the original 8,737 CT scans was annotated with bounding boxes for the left kidney, right kidney, and aorta vessel. The model was trained on this subset and tested on the full dataset to generate annotated output. The training hyperparameters are summarized in Table IV. Then, bounding box coordinates were used to remove all organs outside the annotated regions, preserving the original image size and

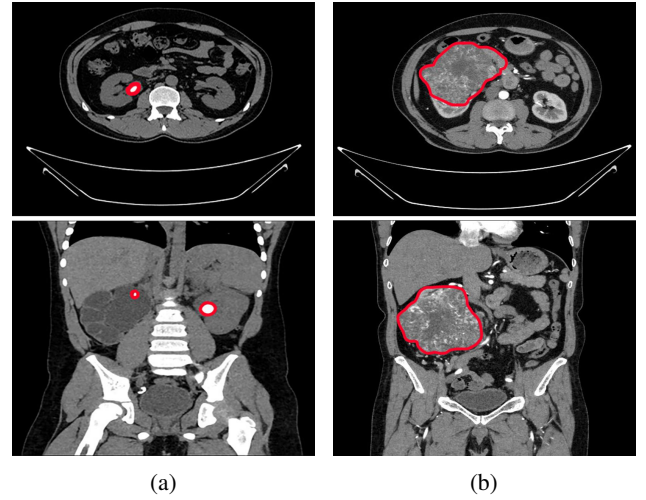


Fig. 3: Manual annotation of CT-scan images under the supervision of a medical expert (nephrologist): (a) Renal calculi, (b) Renal carcinoma.

TABLE IV: Training Hyperparameters for kidney localization

Parameter	Value
Input Image Size	640×640
Batch Size	16
Epochs	80
Learning Rate	0.001667
Optimizer	AdamW

aspect ratio. This background removal enabled ROI reduction, focusing analysis on the regions inside the three bounding boxes, each of which is a rectangular region that localizes and encloses the left kidney, right kidney, and aorta in the CT-scan image. The entire kidney localization workflow is shown in Fig. 6.

2) Contour-Guided Quickshift Segmentation: To efficiently identify regions of interest in CT-scan images, we propose a novel contour-guided quickshift segmentation technique. This method combines unsupervised segmentation with intensity-based contour refinement, resulting in accurate and coherent region grouping. Unlike conventional superpixel-based approaches, our method integrates spatial-color clustering and contour-aware merging, making it more robust for downstream analysis such as object detection and classification.

The first step involves unsupervised superpixel generation using the Quickshift algorithm. This method identifies clusters of pixels based on density peaks in a combined spatial and color feature space. Each pixel's density is estimated using a kernel-weighted function:

$$P_i = \frac{1}{N} \sum_{j=1}^N \phi(D_{ij}), \quad (1)$$

where:

- P_i is the estimated density at pixel i ,
- D_{ij} denotes the distance between pixels i and j in feature space,
- $\phi(D_{ij})$ is the kernel function, typically a Gaussian:

$$\phi(D_{ij}) = \exp\left(-D_{ij}^2/(2\sigma^2)\right), \quad (2)$$

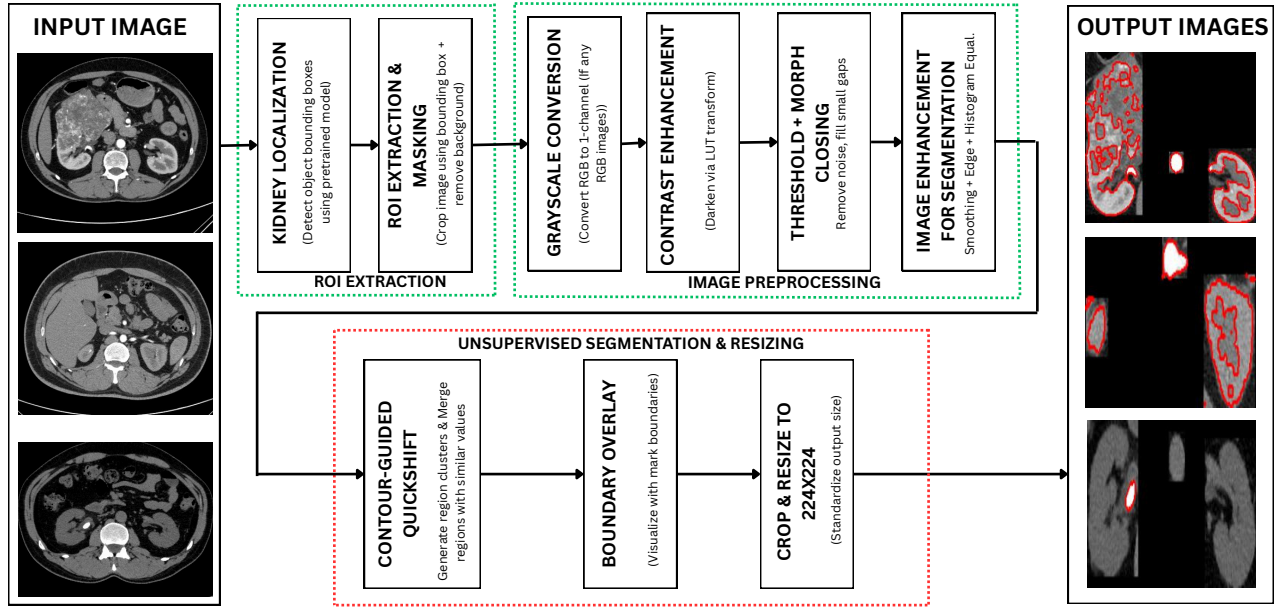


Fig. 4: Top-level overview of the proposed RenSeg for unsupervised segmentation of renal anomalies. CT slices are preprocessed and passed through a contour-guided Quickshift algorithm to generate automated pseudo-labels, which are then used to train a transfer learning network without manual annotation.

- σ is the kernel bandwidth parameter,
- N is the total number of pixels in the image.

Each pixel is then shifted toward its nearest neighbor of higher density to form clusters:

$$y_i(1) = \arg \min_{j: P_j > P_i} D_{ij}, \quad (3)$$

To control over-segmentation, a distance threshold τ is introduced to limit excessive shifting:

$$D_{ij} \leq \tau, \quad (4)$$

After initial segmentation, the resulting superpixels often contain redundant or over-segmented boundaries. To address this, we introduce a contour generalization mechanism that merges adjacent regions based on their intensity similarity. For any two neighboring regions A and B with mean intensities μ_A and μ_B , a merging criterion is defined as:

$$|\mu_A - \mu_B| < \tau, \quad (5)$$

Here, τ is a predefined threshold controlling the merging sensitivity. If the criterion is satisfied, regions are combined, and the average intensity of the merged region is recalculated. This iterative process progressively merges over-segmented areas into semantically meaningful contours.

This contour-guided refinement enhances the interpretability and compactness of the segmentation, making it highly suitable for subsequent tasks such as lesion detection, anatomical localization, and classification. The hyperparameters used are listed in Table V, and the complete contour-guided Quickshift segmentation procedure is summarized in Algorithm 2. A visual representation of the entire segmentation workflow is shown in Fig. 6.

TABLE V: Contour-Guided Quickshift Hyperparameters

Parameter	Value
Kernel Size	3
Ratio (Color Radius)	0.5
Max Distance	8
Bandwidth (Sigma for Gaussian)	1

3) Data Resizing: As part of the RenSeg framework, the segmented CT-scan images containing the ROI are cropped and resized to a standard resolution of $224 \times 224 \times 3$ for uniform input to the classification models. Maximum, average, and minimum image dimensions were analyzed to guide resizing, which was performed using OpenCV in Python. A visual summary of this step within the RenSeg workflow is shown in Fig. 6.

Algorithm 1 Proposed RenSeg Method

Input: CT-scan images (C)

Output: Refined Segmented Images of Kidney and Aorta

```

1: function RENSEG( $C$ )
2:    $subC \leftarrow \text{SELECT\_SUBSET}(C)$ 
3:    $locImg \leftarrow \text{PERFORM\_LOCALIZATION}(subC)$ 
4:    $data \leftarrow \text{CREATE\_DATASET}(locImg)$ 
5:    $segModel \leftarrow \text{TRAIN\_SEGMENTATION\_MODEL}(data)$ 
6:    $coords \leftarrow \text{EXTRACT\_COORDINATES}(segModel, C)$ 
7:    $roiImg \leftarrow \text{EXTRACT\_REGIONS\_OF\_INTEREST}(C, coords)$ 
8:    $normImg \leftarrow \text{NORMALIZE}(roiImg, targetSize)$ 
9:    $grayImg \leftarrow \text{CONVERT\_TO\_GRAYSCALE}(normImg)$ 
10:   $smoothImg \leftarrow \text{GAUSSIAN\_FILTER}(grayImg, \sigma = 1)$ 
11:   $edgeEnh \leftarrow \text{ENHANCE\_EDGES}(smoothImg)$ 
12:   $refinedSeg \leftarrow \text{CONTOUR\_GUIDED\_QUICKSHIFT}(\text{edgeEnh}, \text{kernelSize}, \text{maxDist}, \text{ratio}, \text{threshold})$ 
13:   $resizeImg \leftarrow \text{CROP\_RESIZE}(refinedSeg)$ 
14:  return  $resizeImg$ 
15: end function

```

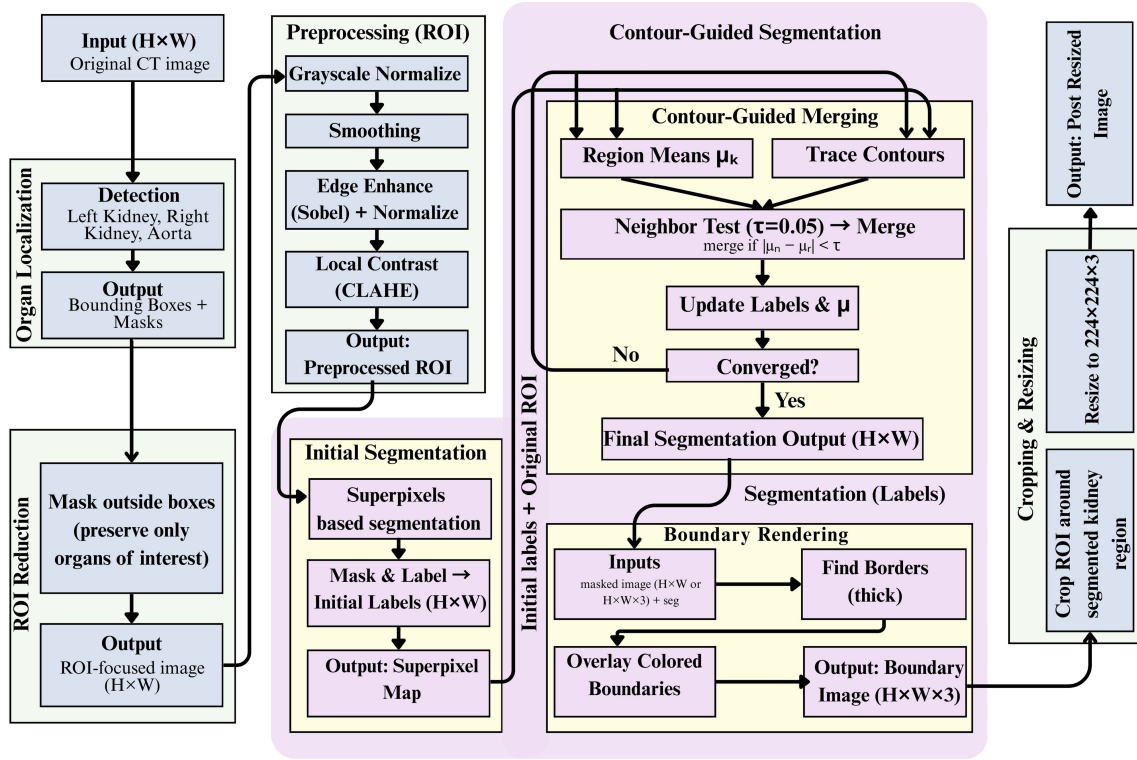


Fig. 5: The Architecture of the Proposed RenSeg Model.

Algorithm 2 Proposed Contour-Guided Quickshift Segmentation**Input:** Localized CT-scan image (I)**Output:** Refined Segmentation Labels

```

1: function CONTOUR_GUIDED_QUICKSHIFT( $I$ )
2:    $segmentation \leftarrow \text{APPLY\_CONTOUR\_GUIDED\_QUICKSHIFT}($ 
3:      $I, \text{kernelSize}, \text{maxDist}, \text{ratio}, \tau)$ 
4:    $labels \leftarrow \text{GET\_UNIQUE\_LABELS}(segmentation)$ 
5:   for each  $label$  in  $labels$  do
6:      $regionA \leftarrow \text{GET\_REGION}(segmentation, label)$ 
7:      $intA \leftarrow \text{MEAN\_INTENSITY}(regionA)$ 
8:     for each  $neighbor$  in  $\text{NEIGHBORS}(regionA)$  do
9:        $regionB \leftarrow \text{GET\_REGION}(segmentation, neighbor)$ 
10:       $intB \leftarrow \text{MEAN\_INTENSITY}(regionB)$ 
11:      if  $|intA - intB| < \tau$  then
12:         $\text{MERGE}(regionA, regionB)$ 
13:      end if
14:    end for
15:  end for
16:  return  $segmentation$ 
17: end function

```

D. Data Splitting

Both the processed dataset by the proposed RenSeg approach and the nephrologist-annotated dataset were split for experimental evaluation. Each dataset comprises a total of 8,737 CT kidney images. To ensure consistent evaluation, both datasets were split into three subsets: training (60%), validation (20%), and testing (20%). This stratified division was applied uniformly to maintain a balanced distribution of samples across the sets. Table VI presents the final distribution of the training, validation, and testing sets.

TABLE VI: Dataset Splitting Summary

	Normal	Calculi	Carcinoma	Total
Training set	3045	825	1369	5239
Validation set	1016	276	457	1749
Testing set	1016	276	457	1749

E. Data Augmentation

This study employed 8,737 CT kidney images across three classes: Normal (5,077), Calculi (1,377), and Carcinoma (2,283). To mitigate class imbalance in the training set, data augmentation was applied, yielding a balanced set of 9,135 images (3,045 per class). Augmentation techniques included flipping, rotation, brightness/contrast adjustment, Gaussian blurring, and gamma correction. It is important to note that the validation and test sets were kept unaugmented to preserve the integrity of the evaluation process and prevent data leakage.

F. Model Selection

Following preprocessing of the RenSeg-generated unsupervised dataset and the manually annotated dataset, this study applied transfer learning models (ResNet50, VGG19, MobileNetV2) to classify normal, carcinoma, and calculi kidneys, leveraging their effectiveness in medical imaging and suitability for limited annotated data.

1) **ResNet50:** ResNet50, as utilized by Zhang et al. [22], is a 50-layer CNN with 25.6M parameters pretrained on ImageNet (14 million images). Its residual connections address the degradation problem in deep networks, enabling effective extraction of discriminative features from biomedical images.

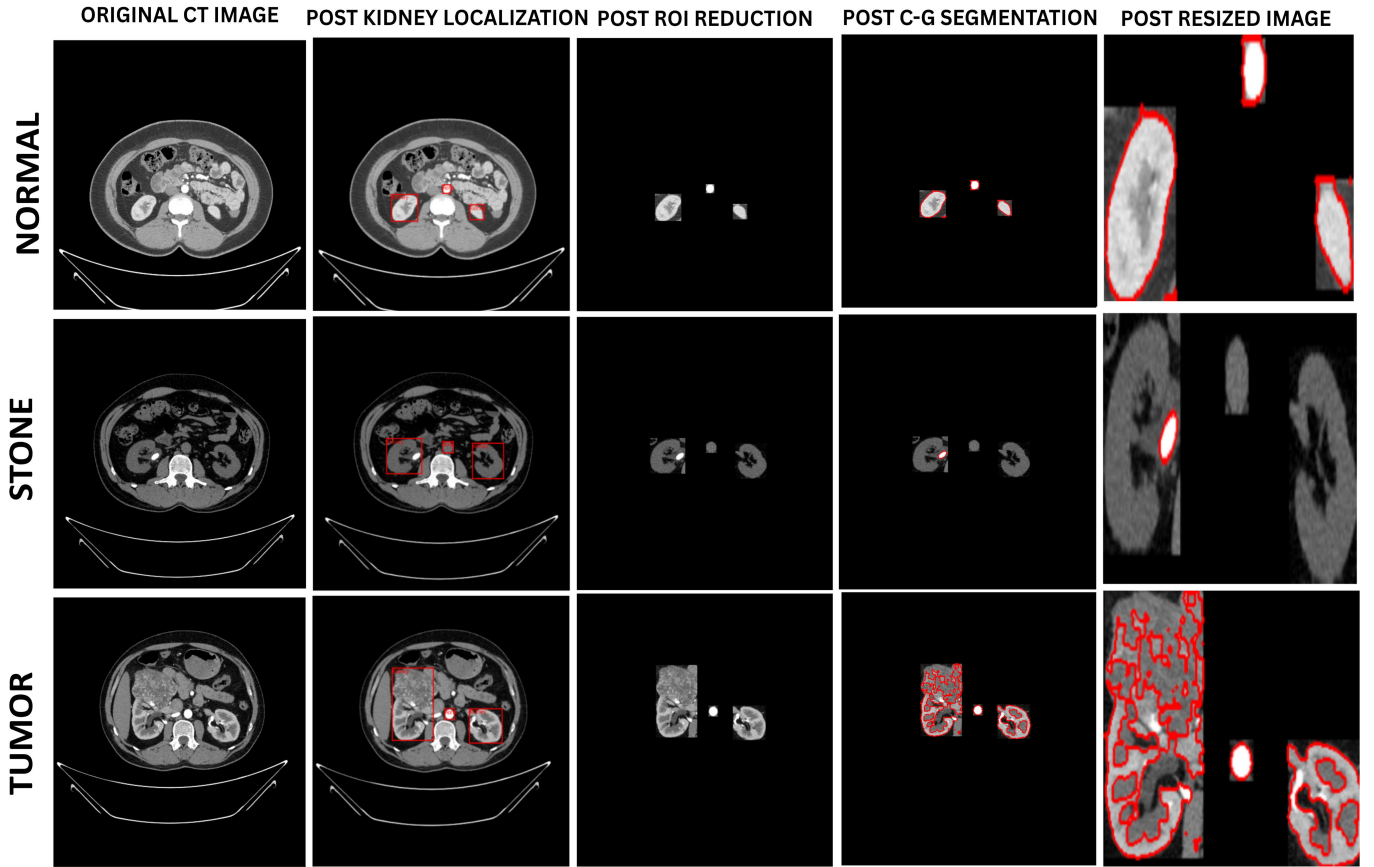


Fig. 6: Sequential pipeline of the RenSeg framework. From left to right: (1) Original axial CT slices; (2) Region of interest (ROI) localization with bounding boxes highlighting kidneys; (3) Cropped ROIs extracted for focused processing; (4) Contour-guided quickshift segmentation for automated labeling and precise boundary extraction; (5) Resizing and standardized output generation.

2) **VGG19**: VGG19, developed by the Visual Geometry Group at Oxford in 2014 [23], is a 19-layer CNN with 143.7M parameters. Its homogeneous 3×3 convolutional design facilitates hierarchical feature extraction and transfer learning. Despite its size, VGG19 demonstrates strong generalization across medical imaging tasks, serving as a deeper comparative baseline.

3) **MobileNetV2**: MobileNetV2, proposed by Al-Gaashani et al. [24], is a lightweight 53-layer CNN with 3.4M parameters. It employs depthwise separable convolutions and inverted residual blocks, enabling efficient inference on resource-constrained devices while maintaining high classification accuracy. This efficiency makes it suitable for real-time clinical decision support systems and mobile applications.

G. Model Fine-Tuning

In this study, pretrained ResNet50, VGG19, and MobileNetV2 models were fine-tuned minimally to adapt to the kidney CT dataset. The convolutional bases were initialized with ImageNet weights and frozen during training to preserve learned features. A custom classification head was added, consisting of a global average pooling layer, a dropout layer for regularization, and dense layers with softmax activation to predict three classes (normal, calculi, carcinoma). The models were compiled with the Adam optimizer and trained

using early stopping and learning rate reduction for stable convergence. This approach enabled efficient transfer learning with low computational cost and reduced overfitting on the limited dataset.

V. RESULT ANALYSIS

This section presents the performance metrics, experimental comparisons, and ablation studies for the proposed RenSeg method.

A. Performance Metrics

In this research, performance metrics like precision, accuracy, recall, and F1-score were used to evaluate classification outcomes. The training history of models was analyzed to assess learning progress, and a confusion matrix provided insight into correct and incorrect predictions, with the corresponding notations given in Table VII. For segmentation, IoU, Dice Coefficient, Specificity, Balanced Accuracy, MCC, HD95, and ASSD were also measured. To ensure statistical reliability across repeated runs, mean accuracy, confidence intervals, and *t*-statistics were computed.

Accuracy (A) measures a model's performance as a percentage or value between 0 and 1, comparing correctly classified samples to total tested samples.

$$A = (a + b) / (a + b + c + d), \quad (6)$$

TABLE VII: Notations for confusion matrix metrics used in formulas

Short Form	Description	Notation
TP	True Positive	a
TN	True Negative	b
FP	False Positive	c
FN	False Negative	d

Recall (R) measures True Positives (TP) out of the total actual positives, aiming to maximize TP without increasing False Negatives (FN).

$$R = a/(a + d), \quad (7)$$

Precision (P) calculates the proportion of true positives (TP) among all predicted positives, reducing the impact of false positives (FP).

$$P = (a)/(a + c), \quad (8)$$

The F1 score is a balanced metric for imbalanced classes, combining precision and sensitivity to reduce both Type I and Type II errors.

$$F_1 = 2 \cdot P \cdot R/(P + R), \quad (9)$$

The Dice Coefficient (DC), or Sørensen's Coefficient, measures overlap between predicted and ground truth pixels in image segmentation.

$$DC = 2|A \cap B|/(|A| + |B|), \quad (10)$$

Intersection over Union (IoU), or Jaccard Index, measures the similarity between predicted and ground truth masks as the ratio of intersection to union of two sets, A (predicted pixels) and B (ground truth pixels). It is defined as:

$$IoU = |A \cap B|/|A \cup B|, \quad (11)$$

Specificity (S) measures the proportion of true negatives correctly identified among all actual negatives.

$$S = b/(b + c), \quad (12)$$

Balanced Accuracy (BA) is the average of recall (sensitivity) and specificity.

$$BA = \frac{1}{2} (a/(a + d) + b/(b + c)), \quad (13)$$

Matthews Correlation Coefficient (MCC) is a correlation coefficient between observed and predicted classifications.

$$MCC = (ab - cd)/\sqrt{(a + c)(a + d)(b + c)(b + d)}, \quad (14)$$

95th Percentile Hausdorff Distance (HD95) measures the boundary error between predicted and ground truth masks, robust to outliers.

$$HD_{95}(A, B) = \text{percentile}_{95} \left(\max_{a \in A} \min_{b \in B} \|a - b\| \right), \quad (15)$$

Average Symmetric Surface Distance (ASSD) measures the average distance between predicted and ground truth boundaries symmetrically.

$$ASSD(A, B) = \frac{1}{|A| + |B|} \left(\sum_{a \in A} \min_{b \in B} \|a - b\| + \sum_{b \in B} \min_{a \in A} \|b - a\| \right), \quad (16)$$

The mean accuracy calculates the average performance across n independent runs.

$$\bar{x} = (1/n) \sum_{i=1}^n x_i, \quad (17)$$

The 95% confidence interval (CI) provides a range within which the true mean performance lies with 95% certainty.

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \cdot (s/\sqrt{n}), \quad (18)$$

where $t_{\alpha/2, n-1}$ is the critical value from the t -distribution, s is the standard deviation, and n is the number of runs.

The t -statistic evaluates the significance of the difference between two models A and B .

$$t = (\bar{x}_A - \bar{x}_B)/\sqrt{s_A^2/n_A + s_B^2/n_B}, \quad (19)$$

B. Experimental Results

The RenSeg approach was evaluated using the same kidney CT dataset, preprocessed through two methods: manual annotation and RenSeg-based segmentation. Both datasets were used to train ResNet50, MobileNetV2, and VGG19 models. Experiments were conducted on a workstation with an AMD Ryzen 9 5950X CPU, 64 GB RAM, and an NVIDIA RTX 3080 Ti GPU. During training, early stopping with a patience of 10 epochs was applied to prevent overfitting. Training was halted if the validation loss did not improve over 10 consecutive epochs, and the model checkpoint with the lowest validation loss was retained for final evaluation.

1) Kidney Localization : The RenSeg technique employed a sub-dataset for kidney localization, with YOLOv8 trained over 80 epochs. As shown in Fig. 7, a steady decrease in box and classification losses indicated improved detection accuracy, while reduced distribution focal loss demonstrated enhanced focus on key features. Specificity remained stable, ensuring reliable true positive identification. Although validation loss declined more slowly, this suggested effective generalization. The validation distribution focal loss closely tracked training loss, reflecting stable learning. Strong AP50 and mAP50-95 scores confirmed consistent detection performance across overlap.

Table VIII presents the confusion matrix for YOLOv8, demonstrating high classification accuracy, particularly for kidneys and aortas. The model correctly identified kidneys 451 times, with only 3 instances misclassified as aortas. Similarly, it accurately detected 229 aortas, with 20 misclassifications. Additionally, the model showed improved performance in identifying background instances, exhibiting negligible classification errors in this category.

Furthermore, as shown in Fig. 8, the comparison between labeled images and YOLOv8 predictions highlights the model's

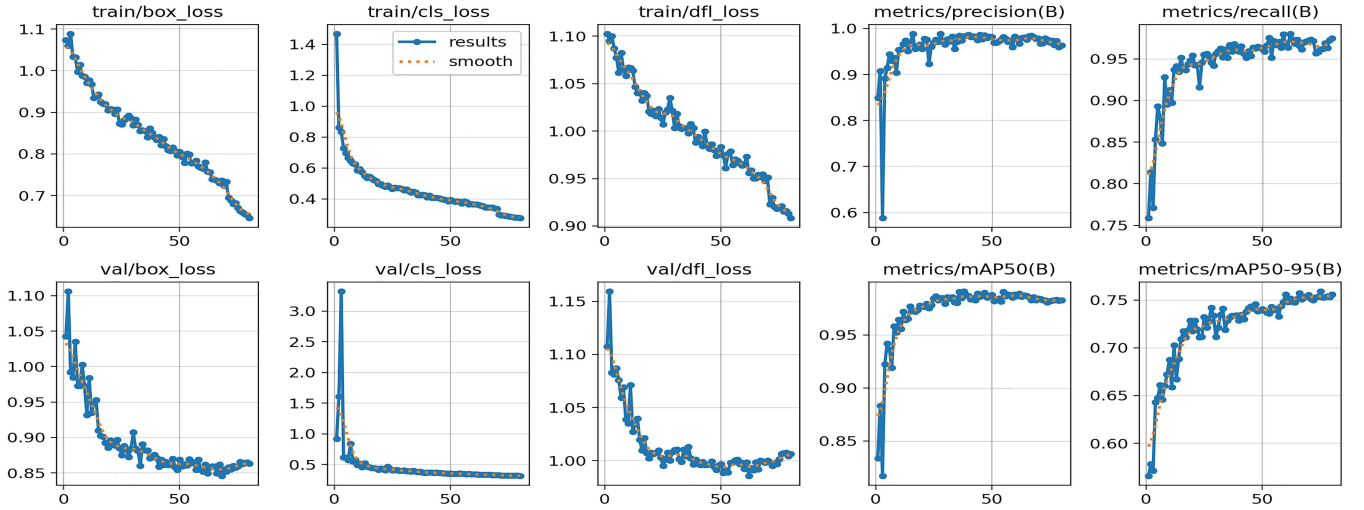


Fig. 7: Loss Curves of YOLOv8 Training and Validation Phases, Including Box, Classification, and DFL Loss

TABLE VIII: Confusion Matrix of YOLOv8

Actual / Predicted	Aorta	Kidney	Background
Aorta	229	0	20
Kidney	1	451	3
Background	4	1	0

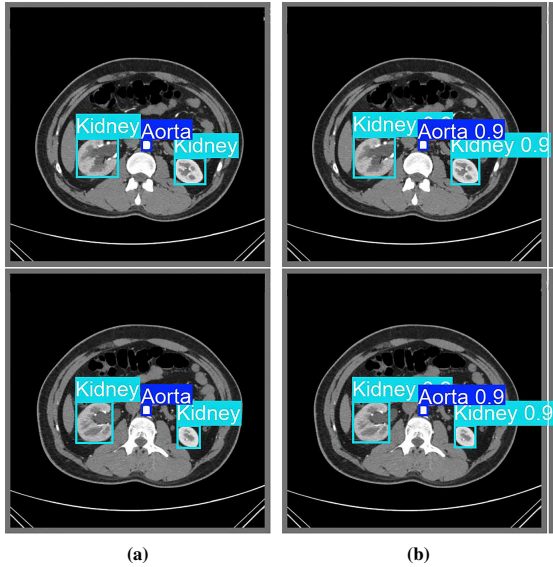


Fig. 8: Comparison between supervised labeled data and model-predicted outputs: (a) Supervised labeled data, (b) Model-predicted outputs.

accuracy in localizing the kidneys and aorta, demonstrating its potential for further reducing cross-testing errors in kidney and aorta identification.

2) *Classification*: The CT-scan image is considered a critical modality in medical imaging for its high-resolution anatomical detail. In this study, both coronal and axial views of the CT-scan image were utilized. As the ROIs were effectively localized and reduced through the RenSeg process, comparable or improved accuracy relative to the manually annotated dataset can be anticipated.

TABLE IX: Classification Performance: RenSeg vs. Manual

Metric	Class	RenSeg			Manual		
		VGG	Res50	MNet	VGG	Res50	MNet
Prec.	Normal	0.96	0.96	0.97	0.89	0.80	0.88
	Calculi	1.00	0.92	0.99	0.99	0.94	1.00
	Carcinoma	0.96	0.93	0.99	0.99	0.74	1.00
Recall	Normal	0.99	0.97	0.99	1.00	0.94	1.00
	Calculi	0.98	0.97	0.99	0.53	0.18	0.50
	Carcinoma	0.91	0.87	0.93	1.00	0.83	0.99
F1	Normal	0.98	0.97	0.98	0.94	0.86	0.93
	Calculi	0.99	0.95	0.99	0.69	0.30	0.67
	Carcinoma	0.94	0.90	0.96	0.99	0.78	0.99
Acc.	—	0.97	0.95	0.98	0.92	0.79	0.92

Table IX summarizes the classification performance of three transfer learning models trained on RenSeg-labeled (unsupervised) and manually annotated (supervised) datasets. Overall, models trained with RenSeg consistently outperformed their manually supervised counterparts across all metrics and classes. MobileNetV2 achieved the highest accuracy (0.98) with RenSeg labeling, while manual annotation resulted in more variable and sometimes lower performance, particularly for the Calculi class. This demonstrates that RenSeg-labeled data can serve as a reliable and scalable alternative to manual annotation for kidney CT-scan image classification.

Table X presents the confusion matrices for VGG19, ResNet50, and MobileNetV2 models using RenSeg-labeled and manually annotated datasets. Each matrix shows classification performance across Normal (N), Calculi (S), and Carcinoma (T) categories. Across all models, predictions with RenSeg are generally more balanced and accurate, particularly for Calculi and Carcinoma. Notably, MobileNetV2 (RenSeg) exhibits the fewest misclassifications, indicating robust feature extraction and class separation. Fig. 9 shows that models trained on RenSeg-labeled data converge more smoothly and consistently than those trained on manual

TABLE X: Confusion matrices of different models using RenSeg and manual annotation. TN = True Normal, TS = True Calculi, TT = True Carcinoma, PN = Predicted Normal, PS = Predicted Calculi, PT = Predicted Carcinoma.

(a) VGG19 (RenSeg)				(b) ResNet50 (RenSeg)				(c) MobileNetV2 (RenSeg)			
TN	1005	0	11	TN	988	5	23	TN	1010	1	5
TS	0	271	5	TS	0	269	7	TS	1	274	1
TT	40	1	416	TT	43	18	396	TT	31	1	425
	PN	PS	PT		PN	PS	PT		PN	PS	PT
(d) VGG19 (Manual)				(e) ResNet50 (Manual)				(f) MobileNetV2 (Manual)			
TN	1016	0	0	TN	950	0	66	TN	1015	1	0
TS	139	138	0	TS	161	50	66	TS	125	146	6
TT	6	0	455	TT	75	3	383	TT	0	0	461
	PN	PS	PT		PN	PS	PT		PN	PS	PT

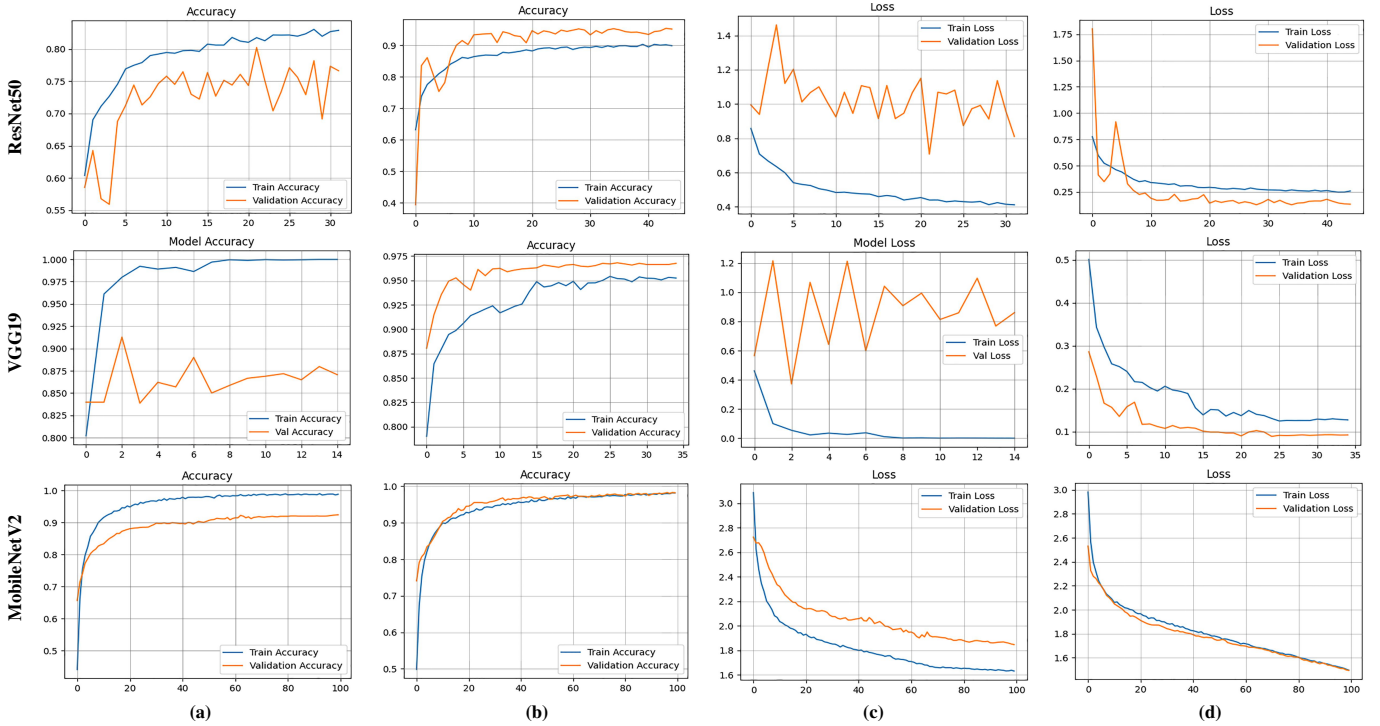


Fig. 9: Training history of models across datasets. Columns correspond to: (a) Accuracy (Manually-annotated), (b) Accuracy (RenSeg), (c) Loss (Manually-annotated), (d) Loss (RenSeg). Rows correspond to models (ResNet50, VGG19, MobileNetV2).

TABLE XI: Comparison of computational complexity in terms of FLOPs (Floating Point Operations) per inference for different classification backbones.

Model	FLOPs (per inference)
MobileNetV2	616,411,090 (~0.616 GFLOPs)
ResNet50	7,756,444,818 (~7.756 GFLOPs)
VGG19	39,038,384,146 (~39.038 GFLOPs)

annotations, which display greater fluctuations in accuracy and loss. RenSeg-based labeling supports uniform, reliable automated annotations, improving generalization and accelerating learning. All models achieved higher performance with RenSeg compared to manual labels, as reflected in macro-average ROC-AUC scores: VGG19 increased from 0.9877 to 0.9912, ResNet50 improved substantially from 0.9194 to

0.9887, and MobileNetV2 attained the highest score, rising from 0.9957 to 0.9979. Table XI presents the floating-point operations (FLOPs) of the classification models trained on RenSeg-processed data. The results indicate that lightweight architectures, such as MobileNetV2, achieve state-of-the-art performance with substantially lower computational cost compared to ResNet50 and VGG19, making MobileNetV2 the most efficient choice for deployment while maintaining strong classification accuracy.

3) Statistical Validation: This subsection provides a statistical validation of model accuracies over 10 independent runs using manual and RenSeg-based automated annotations.

Table XII presents mean accuracy, 95% confidence intervals (CIs), and t -statistics for VGG19, ResNet50, and MobileNetV2, evaluated under manual and RenSeg-based annotation approaches. Models trained with RenSeg-labeled

TABLE XII: Statistical analysis of model performance across 10 independent runs, including mean accuracy with 95% confidence interval margin of error ($\pm$), and t-statistics for VGG19, ResNet50, and MobileNetV2 under manual annotation and the RenSeg-based automated annotation approach.

Model	Mean Accuracy $\pm$ 95% CI Margin	T-Statistic
Manual Annotation		
VGG19	0.8804 ± 0.0222	38.7801
ResNet50	0.6904 ± 0.0458	9.4040
MobileNetV2	0.8254 ± 0.0048	154.8731
RenSeg Annotation		
VGG19	0.9481 ± 0.0038	269.3522
ResNet50	0.8039 ± 0.0499	13.7885
MobileNetV2	0.9089 ± 0.0069	135.5075

data consistently outperformed those trained with manual annotations in both accuracy and stability. RenSeg-VGG19 achieved the highest mean accuracy (0.9481) with a narrow CI (0.9444, 0.9519), indicating superior predictive performance and minimal variance. MobileNetV2 with RenSeg-based annotation reached a mean accuracy of 0.9089 and a high t-statistic (135.51), reflecting strong consistency and robustness. In contrast, ResNet50 under manual annotation recorded the lowest mean accuracy (0.6904) and the widest CI (0.6446, 0.7362), indicating substantial variability and weaker reliability. The *t*-statistics further highlight stability differences: VGG19 with RenSeg achieved 269.35, while ResNet50 with manual annotation had the lowest stability ($t = 9.40$). All models yielded *p*-values below 0.001, confirming that performance improvements with RenSeg-based automated annotation are statistically significant. These results demonstrate that RenSeg enhances both accuracy and reliability, providing a robust, scalable alternative to manual labeling for medical imaging applications.

C. Ablation study

An ablation study was conducted to evaluate the contribution of each stage in the RenSeg model using three datasets: raw CT-scan images, localized kidney regions, and the final RenSeg output with both localization and segmentation. ResNet50, MobileNetV2, and VGG19 were trained on each dataset to classify Normal, Calculi, and Carcinoma cases. As shown in Table XIII, model performance improved after localization and peaked with the full RenSeg model, highlighting the benefit of targeted extraction combined with unsupervised segmentation. Accuracy, macro F1-score, and ROC-AUC consistently increased across all models. For instance, ResNet50's macro F1-score increased from 0.54 (Original) to 0.68 (Post-Localization) and further to 0.94 (RenSeg). Similarly, MobileNetV2 rose from 0.72 to 0.81 and then to 0.98, while VGG19 improved from 0.76 to 0.85 and then to 0.97. Although nephrologist annotation is traditionally considered the gold standard, RenSeg achieves comparable results on CT-scan images with greater efficiency. On the RenSeg-processed dataset, MobileNetV2 attained the highest ROC-AUC (0.9990), ResNet50 showed the largest relative accuracy gain (0.67 to 0.95), and VGG19 consistently performed well

TABLE XIII: Comparative ablation study of ResNet50, MobileNetV2, and VGG19 performance on Original, Post-Localization, and RenSeg-processed datasets.

Dataset	Model	F1 (Macro)	Acc.	ROC-AUC
Original	ResNet50	0.54	0.67	0.8042
	MobileNetV2	0.72	0.72	0.8993
	VGG19	0.76	0.81	0.9506
Post-Localization	ResNet50	0.68	0.75	0.8891
	MobileNetV2	0.81	0.86	0.9032
	VGG19	0.85	0.88	0.9858
RenSeg	ResNet50	0.94	0.95	0.9886
	MobileNetV2	0.98	0.98	0.9990
	VGG19	0.97	0.97	0.9967

(0.97 accuracy, 0.9967 ROC-AUC), demonstrating RenSeg's effectiveness across models for kidney disease detection.

VI. DISCUSSION

Human-annotated datasets are widely regarded as the gold standard for medical imaging. However, the RenSeg technique demonstrates comparable results, particularly in calculi detection. For comparison purposes, both the watershed segmentation method [25], an unsupervised technique commonly used in medical imaging, and CUTS (Contrastive Unpaired Translation and Segmentation) [26], a deep learning and topological framework that leverages contrastive learning and multigranular partitioning to achieve unsupervised medical image segmentation, were employed. In contrast to RenSeg, which processes the entire image, the segmented areas, such as calculi or carcinoma, identified by RenSeg have been reintegrated into the original CT scan for a more accurate comparison.

The qualitative comparison in Fig. 10 shows that RenSeg, Watershed, and CUTS were evaluated against expert nephrologist annotations for both renal calculi and carcinoma cases. RenSeg produced segmentation closely aligned with expert labels, even under the variability in carcinoma shapes and intensities, while Watershed frequently failed to capture precise boundaries and maintain regional consistency. CUTS, despite its unsupervised deep learning design, showed only marginal improvements over Watershed and remained less effective on this dataset. In contrast, RenSeg achieved high-quality segmentation for both calculi and carcinoma, with calculi cases showing particularly accurate alignment due to their distinct structure.

For quantitative validation, RenSeg was evaluated on 1,377 renal calculi and 2,283 carcinoma CT-scan images from the CT kidney dataset [14], alongside Watershed and CUTS. All methods were benchmarked against expert annotations under nephrologist supervision using standard evaluation metrics. As summarized in Table XIV, RenSeg consistently outperformed both competing methods. Unlike Watershed, which often failed to capture precise boundaries or maintain regional consistency, RenSeg handled anatomical variability and carcinoma heterogeneity with notable precision. CUTS, despite its advanced architecture, showed limited effectiveness on this dataset and performed only slightly better than Watershed. These results

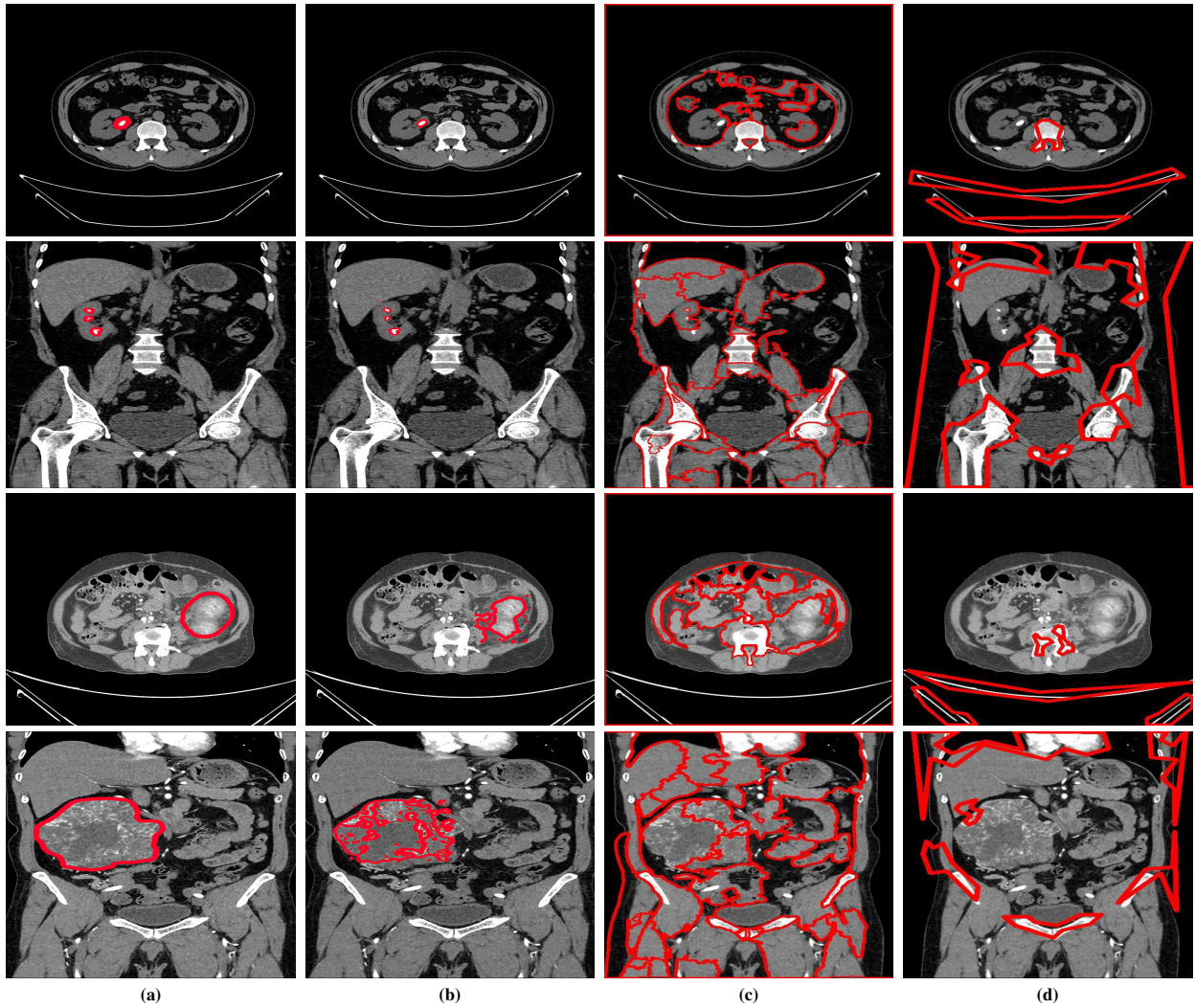


Fig. 10: Comparative visualization of results on CT-scan images for renal calculi and carcinoma. Each row shows a different case, comparing (a) manual annotation (ground truth) with predictions from (b) RenSeg (proposed method), (c) Watershed (unsupervised), and (d) CUTS (unsupervised).

TABLE XIV: Comparison of segmentation performance metrics for RenSeg, Watershed, and CUTS on renal calculi and carcinoma classes.

Method	Class	IoU	Precision	Recall	Dice	Accuracy	Specificity	Balanced Acc	MCC	HD95	ASSD
RenSeg	Calculi (Avg)	0.8997	0.9506	0.9411	0.9458	0.9929	0.9966	0.9688	0.9418	1.1494	0.2702
	Carcinoma (Avg)	0.8738	0.9508	0.9125	0.9309	0.9885	0.9951	0.9538	0.9246	1.6137	0.3815
Watershed	Calculi (Avg)	0.0019	0.0019	0.2130	0.0037	0.8529	0.8536	0.5336	0.0069	448.3009	235.0908
	Carcinoma (Avg)	0.0216	0.0224	0.3855	0.0415	0.8735	0.8776	0.6315	0.0635	416.4881	200.9500
CUTS	Calculi (Avg)	0.0002	0.0002	0.0197	0.0003	0.9181	0.9192	0.4698	-0.0073	394.9902	234.2387
	Carcinoma (Avg)	0.0039	0.0044	0.0353	0.0075	0.9231	0.9300	0.4827	-0.0109	313.5795	163.5236

TABLE XV: Average Empirical execution time comparison (in milliseconds) of RenSeg, Watershed, and CUTS.

Method	Average Execution Time per Image (ms)
RenSeg (Proposed)	60.33
Watershed	88.99
CUTS	732.59

highlight RenSeg’s capability to address the complexities of medical image segmentation.

Furthermore, Table XV reports the average empirical execution times of the three segmentation methods. RenSeg achieved not only the highest segmentation accuracy but also superior computational efficiency, requiring substantially less time than both Watershed and CUTS. Execution times were measured using the hardware configuration specified in Section V-B, ensuring fair comparison across methods. Notably, RenSeg reduced the average execution time to nearly one-third of CUTS while maintaining higher accuracy, underscoring its suitability for time-sensitive clinical applications.

VII. CONCLUSION

This study introduced RenSeg, an unsupervised segmentation-based model for renal calculi and carcinoma analysis from CT-scan images. As highlighted in the discussion, RenSeg achieved superior segmentation and classification performance compared with conventional manual annotation and supervised baselines, as well as unsupervised techniques such as Watershed and CUTS, while the outlined limitations indicate directions for future research. The findings suggest that reducing reliance on annotated datasets can accelerate the deployment of AI-assisted diagnostic solutions in clinical workflows, aligning with the need for scalable and interpretable predictive analytics to support effective chronic disease management.

Although RenSeg demonstrates strong performance, the current evaluation is limited to renal calculi and carcinoma, and broader validation on other kidney pathologies remains necessary. In addition, the study did not explore privacy-preserving deployment strategies, such as federated learning or encrypted computation frameworks. These aspects represent opportunities for extending RenSeg in future research, where future work will focus on expanding to additional renal pathologies, incorporating multimodal clinical data, and embedding interpretability and privacy-preserving mechanisms to enhance its clinical readiness across multi-center environments.

VIII. DATA AND CODE AVAILABILITY

The primary dataset employed in this study, the CT kidney dataset [14], is publicly accessible and has been properly cited in the manuscript. The manually annotated dataset (Mendeley, <https://data.mendeley.com/preview/rxfk6t8f6x?a=05196cf9-92c0-47ba-96c3-f0640f725967>) and the RenSeg code (<https://github.com/sarweralam867/RenSeg.git>) are currently accessible through private repositories and will be made publicly available upon manuscript acceptance to ensure reproducibility.

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